初等概率论

第一次习题课

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知识回顾

知识回顾

■概率空间（样本空间、事件域、概率）

样本空间：每一个概率模型都关联着一个试验，该试验的所有可能结果形成样本空间

事件域：可以分配概率的样本空间的子集，圈定了全部我们关心的事件的范围

概率：分配到事件域上的一个[0,1]之间的函数，表示了我们对这个事件发生的信念

■概率的性质

公理：非负性、归一性、可列可加性

常用性质：有限可加、单调、容斥原理、可列次可加性；单调事件列的极限

补充性质：一般事件列的上下极限

■概率性质常见证明技巧3个：互斥、归纳、求补

知识回顾

- matching problem等经典模型
- 涉及可列交并事件的概率运算，包括单调列的极限运算，e.g.

▶ 定义：事件列的上极限与下极限

▶ 设$\{A_i, i = 1,2,\dots\}$是Ω中的事件列，定义：

1. $\{A_i, i = 1,2,\dots\}$的上极限，记作$\limsup\limits_{n\to\infty} A_n$，定义为：

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k = \{\omega \in \Omega: \omega\text{属于无穷多个}A_i\}$$

2. $\{A_i, i = 1,2,\dots\}$的下极限，记作$\liminf\limits_{n\to\infty} A_n$，定义为：

$$\liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty} A_k = \{\omega \in \Omega: \omega\text{属于所有的}A_i\text{ 除了有限个之外}\}$$

3. 如果$\liminf\limits_{n\to\infty} A_n = \limsup\limits_{n\to\infty} A_n$，则称事件列$\{A_i\}$的极限存在，记作$\lim\limits_{n\to\infty} A_n$.

知识回顾

■ 随机抽样的4种模式

	抽样方式		不同抽样的总数
自含有M个球箱子中的n次抽样	放回	有序	M^n
		无序	$\binom{M+n-1}{n}$
	不放回	有序	P_M^n
		无序	$\binom{M}{n}$

知识回顾

- 条件概率(conditional probability) 设 $(\Omega,\mathcal{F},\mathbb{P})$ 是概率空间，$A,B\in\mathcal{F}$，且$\mathbb{P}(A)>0$，则在事件 A 发生的条件下，事件 B 发生的条件概率定义为：$\mathbb{P}(B|A)=\frac{\mathbb{P}(A\cap B)}{\mathbb{P}(A)}$.

  条件概率依然是一个概率测度；所有关于概率的性质对条件概率都成立

- （乘法公式）设 $(\Omega,\mathcal{F},\mathbb{P})$ 是概率空间，$A_i\in\mathcal{F}$，$i=1,\dots,n$，且 $\mathbb{P}(A_1\cap A_2\cap\cdots\cap A_{n-1})>0$，则

$$\mathbb{P}\left(\bigcap_{i=1}^{n}A_i\right)=\mathbb{P}(A_1)\prod_{i=2}^{n}\mathbb{P}(A_i|A_1\cap A_2\cap\cdots\cap A_{i-1}).$$

- （全概率公式）设 $(\Omega,\mathcal{F},\mathbb{P})$ 是概率空间，$B,A_i\in\mathcal{F}$，$i=1,\dots,n$，且 $\mathbb{P}(A_i)>0$，$\{A_i\}$ 是 Ω 的一个分割，则

$$\mathbb{P}(B)=\mathbb{P}(A_1)\mathbb{P}(B|A_1)+\cdots+\mathbb{P}(A_n)\mathbb{P}(B|A_n).$$

- （Bayes’ Rule）设 $(\Omega,\mathcal{F},\mathbb{P})$ 是概率空间，$B,A_i\in\mathcal{F}$，$i=1,\dots,n$，且 $\mathbb{P}(B)>0$，$\mathbb{P}(A_i)>0$，$\{A_i\}$ 是 Ω 的一个分割，则

$$\mathbb{P}(A_i|B)=\frac{\mathbb{P}(A_i)\mathbb{P}(B|A_i)}{\sum_{j=1}^{n}\mathbb{P}(A_j)\mathbb{P}(B|A_j)},\qquad i=1,\dots,n.$$

- 独立性(independence) 设 $(\Omega,\mathcal{F},\mathbb{P})$ 是概率空间，$A,B\in\mathcal{F}$，如果 $\mathbb{P}(A\cap B)=\mathbb{P}(A)\mathbb{P}(B)$，则称 A 与 B 相互独立，简称独立.

随机变量相关小结

- 给每个可能的试验结果分配一个数。即，一个从样本空间 Ω 到实数轴 $\mathbb{R}$ 的函数。离散、连续均可能。

重点

- **定义：随机变量（random variable, r.v., rv）**

  设 $(\Omega, \mathcal{F})$ 为可测空间，如果 Ω 上的函数 $X(\omega)$满足：对 $\forall x \in R$, $\{\omega | X(\omega) \le x\} \in \mathcal{F}$，则称 $X(\omega)$ 为 $(\Omega, \mathcal{F})$ 上的随机变量，简称**随机变量**.
  通常将随机变量 $X(\omega)$ 简记为 X.

“可分配概率”的要求，是非重点

- **定义：随机变量的相互独立性**

  设 $X_1, \dots, X_n$ 是$(\Omega, \mathcal{F})$ 上的随机变量，如果对任意的实数 $x_1, \dots, x_n$ 都有

  $$\mathbb{P}(X_1 \le x_1, \dots, X_n \le x_n) = \mathbb{P}(X_1 \le x_1) \cdots \mathbb{P}(X_n \le x_n),$$

  称随机变量 $X_1, \dots, X_n$ 相互独立.

重点

作业题目

作业题：概率性质

某商店售卖一种盲盒，每盒内有一个印有24节气之一的书签。试求购买n盒这种盲盒而集齐24节气书签的概率。(Hint:古典概型、定义合适的事件来表达目标事件、可用课上已学的概率性质；写出最终含n的计算式即可，无需化简)

记 $B=\{$购买的 n 盒盲盒中含有全套 24 节气的书签$\}$, $A_i=\{$购买 n 盒盲盒已收集到第 i 个节气的书签$\}$, 则

$$B=\bigcap_{i=1}^{24} A_i, \quad B^c=\bigcup_{i=1}^{24} A_i^c$$

并且

$$P\left(A_i^c\right)=\frac{(24-1)^n}{24^n}=\left(1-\frac{1}{24}\right)^n$$

$$P\left(A_i^c A_j^c\right)=\frac{(24-2)^n}{24^n}=\left(1-\frac{2}{24}\right)^n, \quad i \neq j$$

$$\cdots\cdots\cdots\cdots$$

$$P\left(A_1^c A_2^c \cdots A_{24}^c\right)=\frac{(24-24)^n}{24^n}=0$$

作业题：概率性质

因而

$$
\begin{aligned}
P(B^c) &= P\left(A_1^c \cup A_2^c \cup \cdots \cup A_{24}^c\right) \\
&= \sum_{i=1}^{24} P\left(A_i\right) - \sum_{1 \leqslant i < j \leqslant 24} P\left(A_i^c A_j^c\right) + \cdots - P\left(A_1^c A_2^c \cdots A_{24}^c\right) \\
&= \binom{24}{1}\left(1-\frac{1}{24}\right)^n - \binom{24}{2}\left(1-\frac{2}{24}\right)^n + \cdots + \binom{24}{23}\left(1-\frac{23}{24}\right)^n \\
&= \sum_{k=1}^{24}(-1)^{k-1}\binom{24}{k}\left(1-\frac{k}{24}\right)^n
\end{aligned}
$$

于是

$$
\begin{aligned}
P(B) &= 1 - P(B^c) \\
&= 1 - \sum_{k=1}^{24}(-1)^{k-1}\binom{24}{k}\left(1-\frac{k}{24}\right)^n \\
&= \sum_{k=0}^{24}(-1)^{k}\binom{24}{k}\left(1-\frac{k}{24}\right)^n
\end{aligned}
$$

作业题：概率模型

Six cups and saucers come in pairs: there are two cups and saucers which are red, two white, and two with stars on. If the cups are placed randomly onto the saucers (one each), find the probability that no cup is upon a saucer of the same pattern.

我们将茶托按顺序排列，简记为 $RRWWSS$。

现考虑同种颜色的茶托和杯子之间也存在差异。将杯子依次放入时，如果要满足题目的要求，那么对于 RR 共有两类情况：

1. 所放置的茶托同色（W_1W_2, S_1S_2）。以 WW 为例，那么余下的杯子 SS 只能放置在茶托 RR 上;
2. 所放置的茶托异色（W_1S_1, W_1S_2, W_2S_1, W_2S_2）。那么放置在茶托 RR 上的杯子只能是 WS 组合。

作业题：概率模型

杯子 RR 放置的茶托 → 同色 $(C_2^1 * A_2^2)$ → SS 放置在 RR 上 (A_2^2) → WW 放置在 SS 上 (A_2^2)

杯子 RR 放置的茶托 → 异色 $(C_2^1 * C_2^1 * A_2^2)$ → WS 放置在 RR 上 $(C_2^1 * C_2^1 * A_2^2)$ → WS 放置在 WS 上 (1)

综上所述，符合要求的放置数为 $C_2^1 * A_2^2 * A_2^2 * A_2^2 + C_2^1 * A_2^2 * C_2^1 * C_2^1 * A_2^2 = 80$ 种。

所求概率为 $80/6! = 1/9$。

作业题

Problem 15. A coin is tossed twice. Alice claims that the event of two heads is at least as likely if we know that the first toss is a head than if we know that at least one of the tosses is a head. Is she right? Does it make a difference if the coin is fair or unfair? How can we generalize Alice's reasoning?

解 2.1： 设事件 A 表示"第一次投掷为正面（Head）"，事件 B 表示"第二次投掷为正面"。题目要求比较

$$\mathbb{P}(A\cap B\mid A) \quad 与 \quad \mathbb{P}(A\cap B\mid A\cup B).$$

首先有

$$\mathbb{P}(A\cap B\mid A)=\frac{\mathbb{P}\big((A\cap B)\cap A\big)}{\mathbb{P}(A)}=\frac{\mathbb{P}(A\cap B)}{\mathbb{P}(A)},$$

以及

$$\mathbb{P}(A\cap B\mid A\cup B)=\frac{\mathbb{P}\big((A\cap B)\cap(A\cup B)\big)}{\mathbb{P}(A\cup B)}=\frac{\mathbb{P}(A\cap B)}{\mathbb{P}(A\cup B)}.$$

由于 $\mathbb{P}(A\cup B)\geq\mathbb{P}(A)$，故

$$\mathbb{P}(A\cap B\mid A)=\frac{\mathbb{P}(A\cap B)}{\mathbb{P}(A)}\geq\frac{\mathbb{P}(A\cap B)}{\mathbb{P}(A\cup B)}=\mathbb{P}(A\cap B\mid A\cup B).$$

因此，Alice 的说法成立，且与硬币是否公平无关。

作业题

Problem 15. A coin is tossed twice. Alice claims that the event of two heads is at least as likely if we know that the first toss is a head than if we know that at least one of the tosses is a head. Is she right? Does it make a difference if the coin is fair or unfair? How can we generalize Alice's reasoning?

公平硬币的数值例： 若硬币公平，则四个结果 HH, HT, TH, TT 等可能。于是

$$\frac{\mathbb{P}(A\cap B)}{\mathbb{P}(A)} = \frac{1/4}{1/2} = \frac{1}{2}, \qquad \frac{\mathbb{P}(A\cap B)}{\mathbb{P}(A\cup B)} = \frac{1/4}{3/4} = \frac{1}{3}.$$

可见在已知“第一次为正面”的条件下得到“两次均正面”的概率为 1/2，而在已知“至少一次为正面”的条件下为 1/3，前者更大。

推广（Alice 推理的抽象形式）： 设 A, B, C 为事件，且满足 $B \subseteq C$ 且 $A\cap B = A\cap C$（例如当 $A \subseteq B \subseteq C$ 时）。则

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A\cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A\cap C)}{\mathbb{P}(B)} \geq \frac{\mathbb{P}(A\cap C)}{\mathbb{P}(C)} = \mathbb{P}(A \mid C),$$

其中不等式由 $\mathbb{P}(C) \geq \mathbb{P}(B)$ 得到。Alice 的情形对应特例 $C = A\cup B$。

作业题：概率不等式

- Bonferroni’s inequality 不等式：

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) \geq \sum_{1\leq i\leq n} \mathbb{P}(A_i) - \sum_{1\leq i<j\leq n} \mathbb{P}(A_i \cap A_j).$$

- 对比容斥恒等式：

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{1\leq i\leq n} \mathbb{P}(A_i) - \sum_{1\leq i<j\leq n} \mathbb{P}(A_i \cap A_j) + \sum_{1\leq i<j<k\leq n} \mathbb{P}(A_i \cap A_j \cap A_k) - \ldots + (-1)^{n-1}\mathbb{P}(A_1 \cap A_2 \cap \ldots \cap A_n)$$

- 直观：多去除了重叠的部分，概率变小了

作业题：概率不等式

- Bonferroni’s inequality 不等式：

$$\mathbb{P}\Big(\bigcup_{i=1}^{n} A_i\Big) \geq \sum_{1\leq i\leq n} \mathbb{P}(A_i) - \sum_{1\leq i<j\leq n} \mathbb{P}(A_i \cap A_j).$$

Bonferroni’s inequality: When n = 2, we have

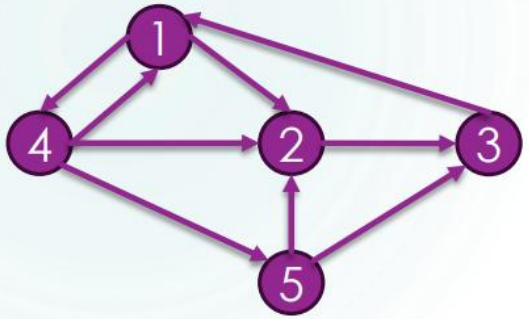
$$\mathbf{P}(A_1 \cup A_2) = \mathbf{P}(A_1) + \mathbf{P}(A_2) - \mathbf{P}(A_1 \cap A_2).$$

Assuming that the inequality holds when n = k-1, then when n = k, we have

$$\begin{aligned}
\mathbf{P}\left(\bigcup_{i=1}^{k} A_i\right) &= \mathbf{P}\left(A_k \bigcup \left(\bigcup_{i=1}^{k-1} A_i\right)\right) \\
&= \mathbf{P}\left(A_k\right) + \mathbf{P}\left(\bigcup_{i=1}^{k-1} A_i\right) - \mathbf{P}\left(\bigcup_{i=1}^{k-1}\left(A_k \bigcap A_i\right)\right) \\
&\geq \mathbf{P}\left(A_k\right) + \Sigma_{i=1}^{k-1}\mathbf{P}\left(A_i\right) - \Sigma_{1\leq i<j\leq k-1}\mathbf{P}\left(A_i \bigcap A_j\right) - \Sigma_{i=1}^{k-1}\mathbf{P}\left(A_i \bigcap A_k\right) \\
&= \Sigma_{i=1}^{k}\mathbf{P}\left(A_i\right) - \Sigma_{1\leq i<j\leq k}\mathbf{P}\left(A_i \bigcap A_j\right).
\end{aligned}$$

作业题

(PageRank) The World Wide Web is a collection of linked web pages. We can use a graph to represent these pages and their links: the nodes of the graph are pages $\mathcal{A}$ and there is a directed edge from i to j if page i has a link to j. We can define the rank of page i as $\pi(i) = \sum_{j\in\mathcal{A}} \pi(j)P(j,i)$, where $P(j,i)$ is the fraction of links in j that point to i and is zero if there is no such link. (*The basic idea of the algorithm is due to Larry Page, hence the name PageRank. Since it ranks pages, the name is doubly appropriate*). For convenience, we further restrict $\sum_{j\in\mathcal{A}} \pi(j) = 1$. Please solve $\pi(i), i = 1, \dots, 5$ in the following graph.



For example, we have $P(3,2) = 1, P(4,2) = \frac{1}{3}, P(5,2) = 0$, and $\pi(1) = \pi(2)\cdot 0 + \pi(3)\cdot 1 + \pi(4)\cdot\frac{1}{3} + \pi(5)\cdot 0$ in this graph.

作业题

The balance equations for this graph are

$$\pi(1) = \pi(2)\cdot 0 + \pi(3)\cdot 1 + \pi(4)\cdot \frac{1}{3} + \pi(5)\cdot 0$$
$$\pi(2) = \pi(1)\cdot \frac{1}{2} + \pi(3)\cdot 0 + \pi(4)\cdot \frac{1}{3} + \pi(5)\cdot \frac{1}{2}$$
$$\pi(3) = \pi(1)\cdot 0 + \pi(2)\cdot 1 + \pi(4)\cdot 0 + \pi(5)\cdot \frac{1}{2}$$
$$\pi(4) = \pi(1)\cdot \frac{1}{2} + \pi(2)\cdot 0 + \pi(3)\cdot 0 + \pi(5)\cdot 0$$
$$\pi(5) = \pi(1)\cdot 0 + \pi(2)\cdot 0 + \pi(3)\cdot 0 + \pi(4)\cdot \frac{1}{3}.$$

Solving these equations, we obtain

$$\pi(1) : \pi(2) : \pi(3) : \pi(4) : \pi(5) = 12 : 9 : 10 : 6 : 2.$$

By the restriction $\sum_{i=1}^{5} \pi(i) = 1$, we have

$$\big(\pi(1), \pi(2), \pi(3), \pi(4), \pi(5)\big) = \frac{1}{39}(12, 9, 10, 6, 2).$$

作业题：贝叶斯公式

3. 【是失职，还是无辜？】2017年贵州航天医院三名从事尘肺病诊断的医生以“失职罪”被逮捕。公安机关调查2012-2016年3名医生共经手10708人胸片检查，针对诊断为尘肺病病人的1640份档案，对其中547例进行了重新复核，只查出42例有尘肺病。而据了解，一般和他们经验相当的一位医生独立读片的误判率为22%（即真实没有尘肺病、但却误判为尘肺病的概率）；在实际工作中，这三位医生独立地读片，当有两位或以上的医生认为有病就判为有病。因此当地警方认定他们对尘肺病的“误诊率超过90%”，“远超常规误判率”，造成3000万社保资金流失，以涉嫌“国有事业单位人员失职罪”予以羁押。

我们记事件“病人真实有病”为 A，事件“病人被这三位医生诊断为有病”为 B。那么根据题目背景，我们知道：

1. 病人的真实发病率 $\mathbf{P}(A)$ 近似为 $(42/547) * 1640/10708 = 1.176\%$.
2. 如果病人真实有病，则三位医生一定会诊断为有病，即 $\mathbf{P}(B \mid A) = 1$, $\mathbf{P}(AB) = \mathbf{P}(A)$.
3. 如果病人真实没病，那么在每个医生有接近 $p = 22\%$ 的误判率的情况下，三位医生医生中至少有两位误判的概率 $\mathbf{P}(B \mid A^c) = p^3 + C_3^1 p^2(1-p)$.

综上所述，根据 Bayes 公式，我们有

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(AB)}{\mathbf{P}(B)} = \frac{\mathbf{P}(AB)}{\mathbf{P}(B \mid A)\mathbf{P}(A) + \mathbf{P}(B \mid A^c)\mathbf{P}(A^c)} = \frac{\mathbf{P}(A)}{\mathbf{P}(A) + (1-\mathbf{P}(A))(p^3 + C_3^1 p^2(1-p))}.$$

代入数据计算可知，在被这三位医生诊断为有病的人中，其真实有病的概率 $\mathbf{P}(A \mid B) = 8.76\%$，非常接近公安机关给出的数据 $42/547 = 7.68\%$。因此可以认为医生没有失职。

在这个问题中，要注意每位医生的误判率 p 和最终被诊断为有病但实际没病的误诊率 $\mathbf{P}(A^c \mid B)$ 是两个不同的概念。导致误诊率 $\mathbf{P}(A^c \mid B)$ 过高的主要原因是疾病本身的发病率 $\mathbf{P}(A)$ 就很低。比如对同样的误判率 $p = 22\%$，如果发病率 $\mathbf{P}(A) = 0.2$，那么最终的误诊率 $\mathbf{P}(A^c \mid B)$ 会只有 33.14%.

作业题：考察随机变量的定义

随机变量：

并不是说$\mathcal{F}$只包含这两个事件

Let $\Omega = \{1,2,3,4,5,6\}$, and let $\mathcal{F} = \sigma(\{1,2,3,4\},\{3,4,5,6\})$. Is the function

$$X(\omega) = 2, \qquad \omega = 1,2,3,4$$

$$X(\omega) = 7, \qquad \omega = 5,6$$

a random variable over $(\Omega, \mathcal{F})$?

定义：随机变量（random variable, r.v., rv）

设 $(\Omega, \mathcal{F})$ 为可测空间，如果 Ω 上的函数 $X(\omega)$满足：对 $\forall x \in R$, $\{\omega | X(\omega) \le x\} \in \mathcal{F}$，则称 $X(\omega)$ 为 $(\Omega, \mathcal{F})$ 上的随机变量，简称**随机变量**.
通常将随机变量 $X(\omega)$ 简记为 X.

$$x < 2, \{\omega | X(\omega) \le x\} = \emptyset \in \mathcal{F}$$

$$2 \le x < 7, \{\omega | X(\omega) \le x\} = \{1,2,3,4\} \in \mathcal{F}$$

$$x \ge 7, \{\omega | X(\omega) \le x\} = \{1,2,3,4,5,6\} \in \mathcal{F}$$

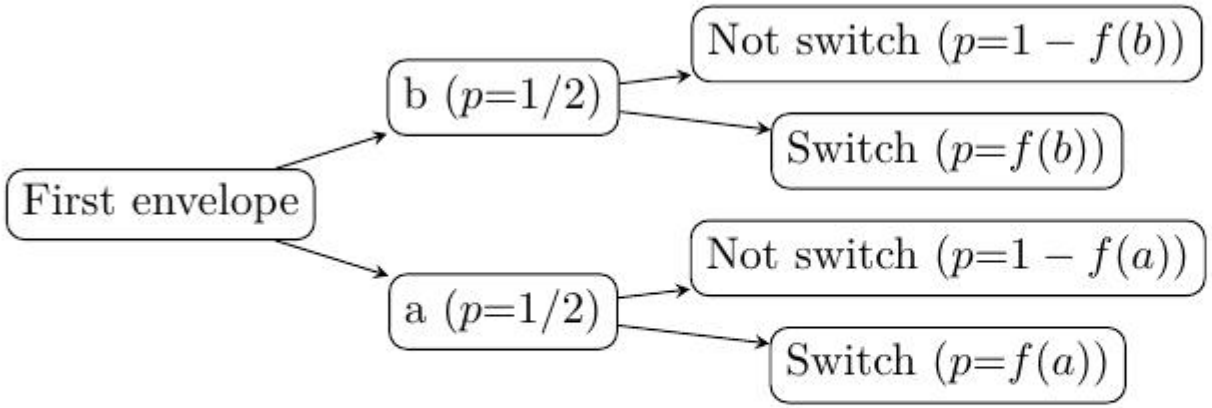
补充题目

补充题：概率性质

Problem 25. A two-envelopes puzzle. You are handed two envelopes, and you know that each contains a positive integer dollar amount and that the two amounts are different. The values of these two amounts are modeled as constants that are unknown. Without knowing what the amounts are, you select at random one of the two envelopes, and after looking at the amount inside, you may switch envelopes if you wish. A friend claims that the following strategy will increase above 1/2 your probability of ending up with the envelope with the larger amount: toss a coin repeatedly, let X be equal to 1/2 plus the number of tosses required to obtain heads for the first time, and switch if the amount in the envelope you selected is less than the value of X. Is your friend correct?

补充题：概率性质

Suppose the amounts are a and b, $a > b$.



Then the probability of selecting a is $1/2 + 1/2(f(b) - f(a))$. It is more than $1/2$ if and only if $f(a) < f(b)$.

In this question, $f(x) = (1/2)^x$ and is monotonically decreasing.

补充题：概率性质

Two interesting questions:

- How can this strategy work?
- Can we devise a method to increase the probability of selecting a larger amount?

补充题：交换悖论

Two envelopes problem (Exchange Paradox): A person is given two indistinguishable envelopes, each of which contains a sum of money. One envelope contains twice as much as the other. The person may pick one envelope and keep whatever amount it contains. They pick one envelope at random but before they open it they are given the chance to take the other envelope instead.

补充题：交换悖论

1. Denote by A the amount in the player's selected envelope.
2. The probability that A is the smaller amount is 1/2, and that it is the larger amount is also 1/2.
3. The other envelope may contain either $2A$ or $A/2$.
4. If A is the smaller amount, then the other envelope contains $2A$.
5. If A is the larger amount, then the other envelope contains $A/2$.
6. Thus the other envelope contains $2A$ with probability 1/2 and $A/2$ with probability 1/2.
7. So the expected value of the money in the other envelope is

$$\frac{1}{2}(2A) + \frac{1}{2}\left(\frac{A}{2}\right) = \frac{5}{4}A$$

8. This is greater than A so, on average, the person reasons that they stand to gain by swapping.
9. After the switch, denote that content by B and reason in exactly the same manner as above.
10. The person concludes that the most rational thing to do is to swap back again.
11. The person will thus end up swapping envelopes indefinitely.
12. As it is more rational to just open an envelope than to swap indefinitely, the player arrives at a contradiction.

补充题：交换悖论

A classical resolution:

Suppose that the total amount in both envelopes is a constant $c = 3x$, with x in one envelope and $2x$ in the other. If you select the envelope with x first you gain the amount x by swapping. If you select the envelope with $2x$ first you lose the amount x by swapping. So you gain on average $G = \frac{1}{2}(x) + \frac{1}{2}(-x) = \frac{1}{2}(x - x) = 0$ by swapping.

So on this supposition that the total amount is fixed, swapping is not better than keeping. The expected value $\mathrm{E} = \frac{1}{2}2x + \frac{1}{2}x = \frac{3}{2}x$ is the same for both the envelopes. Thus no contradiction exists.

补充题：二孩问题

(Part 1) Consider a family that has two children. we are interested in the children ‘s genders. Our sample space is $S = \{(G,G),(G,B),(B,G),(B,B)\}$. Also assume that all four possible outcomes are equally likely. (Note: Part 1无需Bayes法则)

a. What is the probability that both children are girls given that the first child is a girl?

b. We ask the father: "Do you have at least one daughter?" He responds "Yes!" Given this extra information, what is the probability that both children are girls? In other words, what is the probability that both children are girls given that we know at least one of them is a girl?

(Part 2) Here is another variation of the family-with-two-children problem. A family has two children. We ask the father, "Do you have at least one daughter named Lilia?" He replies, "Yes!" What is the probability that both children are girls? In other words, we want to find the probability that both children are girls, given that the family has at least one daughter named Lilia. Here you can assume that if a child is a girl, her name will be Lilia with probability $a \ll 1$ independently from other children's names. If the child is a boy, his name will not be Lilia. Compare your result with **(Part 1)** (b).

补充题：二孩问题

Solution: 1. Conditioning on the first child being a girl, the possible outcomes are

$$\{(G,G),(G,B)\}.$$

Among these 2 equally likely cases, only 1 has both children as girls. Therefore,

$$\mathbb{P}(\text{both girls} \mid \text{first is girl}) = \tfrac{1}{2}.$$

2. Conditioning on the event "at least one child is a girl," the possible outcomes are

$$\{(G,G),(G,B),(B,G)\}.$$

Among these 3 equally likely cases, only 1 has both children as girls. Therefore,

$$\mathbb{P}(\text{both girls} \mid \text{at least one is girl}) = \tfrac{1}{3}.$$

补充题：二孩问题

Solution: For one child, the outcomes and probabilities are

$$\mathbb{P}(G_L) = \tfrac{1}{2}a, \qquad \mathbb{P}(G_{\neg L}) = \tfrac{1}{2}(1-a), \qquad \mathbb{P}(B) = \tfrac{1}{2}.$$

Let A be the event "at least one daughter named Lilia", and G the event "both children are girls".

Denominator. The probability that at least one child is G_L is

$$\mathbb{P}(A) = 1 - \mathbb{P}(\text{no } G_L) = 1 - \left(1 - \tfrac{a}{2}\right)^2 = a - \frac{a^2}{4}.$$

Numerator. Given both children are girls, the chance that at least one is named Lilia is $1-(1-a)^2 = 2a - a^2$. Since $\mathbb{P}(G) = \frac{1}{4}$,

$$\mathbb{P}(G \cap A) = \tfrac{1}{4}(2a - a^2) = \frac{a}{2} - \frac{a^2}{4}.$$

Therefore

$$\mathbb{P}(G \mid A) = \frac{\mathbb{P}(G \cap A)}{\mathbb{P}(A)} = \frac{\frac{a}{2} - \frac{a^2}{4}}{a - \frac{a^2}{4}} = \frac{\frac{1}{2} - \frac{a}{4}}{1 - \frac{a}{4}} = \boxed{\frac{2-a}{4-a}}.$$

Since $a \ll 1$, this is close to $\frac{1}{2}$:

$$\frac{2-a}{4-a} = \frac{1}{2} - \frac{a}{8} + O(a^2) \approx \frac{1}{2}.$$

In Part 1(b), conditioning only on "at least one child is a girl" gives $\frac{1}{3}$. Thus specifying a (rare) name yields much stronger information, pushing the probability from $\frac{1}{3}$ up toward $\frac{1}{2}$ as $a \to 0$.